Azotemia — The 7-Clue Renal Triage

1. Acute renal failure (door 1)

WHAT YOU SEE
Door 1 with an alarm clock — fast onset.

WHAT IT ENCODES
AKI = rise in creatinine over hours-days (KDIGO: Cr up >=0.3 mg/dL in 48h, or 1.5x baseline in 7d, or urine output <0.5 mL/kg/h x6h). Always triage prerenal vs intrinsic vs postrenal first. Reversible if caught early.

BOARD TRAP
Don't anchor on CKD: a known baseline Cr that suddenly jumps is acute-on-chronic, still worked up as AKI.

2. Hematuria / RBC casts / ATN (door 1 placard)

WHAT YOU SEE
Bloody floor under door 1 — hematuria, RBC casts, ATN.

WHAT IT ENCODES
Urine sediment is the cheapest discriminator: RBC casts = glomerulonephritis; muddy-brown granular casts = ATN; WBC casts = interstitial nephritis/pyelo. ATN is the most common intrinsic AKI in hospital.

BOARD TRAP
RBC casts are NOT seen in prerenal azotemia — their presence rules out a pure prerenal cause and points intrinsic.

3. Acute nephritis (door 2 — knight)

WHAT YOU SEE
Knight in armor — acute nephritic syndrome.

WHAT IT ENCODES
Nephritic = hematuria + RBC casts + HTN + mild-moderate proteinuria (<3.5 g) + oliguria. Think PSGN, IgA, anti-GBM, vasculitis. Active urine sediment is the hallmark.

BOARD TRAP
Nephritic proteinuria stays sub-nephrotic; >3.5 g/day pushes you toward nephrotic syndrome instead.

4. Chronic renal failure (door 3)

WHAT YOU SEE
X-ray of small kidneys behind door 3.

WHAT IT ENCODES
CKD = GFR <60 for >=3 months or structural damage. Small echogenic kidneys on US, anemia, hyperphosphatemia, hypocalcemia and high PTH all suggest chronicity rather than acute injury.

BOARD TRAP
Bilateral small kidneys = chronic and irreversible; normal/large kidneys with AKI can still be reversible (e.g., diabetic nephropathy, amyloid, obstruction).

5. Nephrotic syndrome (door 4 — obese man)

WHAT YOU SEE
Edematous man at door 4 — nephrotic.

WHAT IT ENCODES
Nephrotic = proteinuria >3.5 g/day + hypoalbuminemia + edema + hyperlipidemia. Bland sediment with oval fat bodies/'maltese cross'. Hypercoagulable (renal vein thrombosis, loss of antithrombin III).

BOARD TRAP
Frothy urine + heavy proteinuria + lipiduria = nephrotic, not nephritic; absence of RBC casts is the discriminator.

6. GFR gauge (universal clue)

WHAT YOU SEE
Big speedometer reading 'low GFR'.

WHAT IT ENCODES
GFR is the universal staging number across every renal box. Estimated by CKD-EPI/MDRD; creatinine lags behind true GFR. A 50% nephron loss can occur before Cr leaves the normal range.

BOARD TRAP
A 'normal' creatinine does NOT exclude reduced GFR in low-muscle-mass or elderly patients — use eGFR, not raw Cr.

7. Electrolyte / acid-base panel

WHAT YOU SEE
K+, Na+, HCO3- chart — universal clue.

WHAT IT ENCODES
Renal failure drives hyperkalemia, metabolic acidosis (high anion gap from retained acids), hyperphosphatemia and hypocalcemia. These complications, not the Cr number, are what kill or trigger dialysis.

BOARD TRAP
Emergent dialysis indications (AEIOU): refractory Acidosis, Electrolytes (K+), Intoxications, Overload, Uremia — not the absolute Cr value alone.

8. Renal tubular defects (box 6)

WHAT YOU SEE
Tubule diagram — tubular disorders.

WHAT IT ENCODES
Tubular defects = RTAs and Fanconi. Type 1 (distal): can't acidify urine, urine pH >5.5, hypokalemia, stones. Type 2 (proximal): bicarb wasting, Fanconi. Type 4: hypoaldosteronism, hyperkalemia, acidic urine.

BOARD TRAP
Type 4 RTA is the ONLY RTA with hyperkalemia — pairs with diabetic hyporeninemic hypoaldosteronism; the others are hypokalemic.

9. Blood pressure cuff / hypertension (box 7)

WHAT YOU SEE

BP cuff — renovascular/HTN box.

WHAT IT ENCODES

Renovascular disease (RAS) and hypertensive nephrosclerosis. ACEi/ARB started in bilateral RAS causes an acute GFR drop (efferent dilation). FMD in young women, atherosclerotic RAS in older smokers.

BOARD TRAP

A >30% creatinine bump after starting ACEi/ARB suggests bilateral renal artery stenosis — stop and image, don't just push the dose.

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10. Nephrolithiasis (box 8)

WHAT YOU SEE

Man clutching flank — kidney stones.

WHAT IT ENCODES

Calcium oxalate stones most common; non-contrast CT is the gold standard. Struvite (staghorn) from urease-positive Proteus, uric acid stones radiolucent and pH-dependent, cystine in children.

BOARD TRAP

Radiolucent stone on plain film = uric acid (treat by alkalinizing urine) or cystine — not calcium, which is radiopaque.

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11. Urinary tract obstruction (box 9)

WHAT YOU SEE

Distended bladder/post-renal sign.

WHAT IT ENCODES

Postrenal azotemia: obstruction must be bilateral (or unilateral in a solitary kidney) to raise Cr. BPH, malignancy, stones. Hydronephrosis on US; post-void residual and bladder scan are quick screens.

BOARD TRAP

Post-obstructive diuresis after relieving obstruction can cause massive volume/electrolyte loss — monitor and replace, don't ignore the urine output.

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12. Chief Nephrologist + clipboard

WHAT YOU SEE

Doctor at the desk holding the triage clipboard.

WHAT IT ENCODES

The triage algorithm: history + exam, urinalysis with sediment, basic chemistries/eGFR, urine electrolytes (FENa), and renal ultrasound before anything fancy. These 7 clues sort every azotemic patient.

BOARD TRAP

FENa <1% = prerenal, >2% = ATN — but FENa is unreliable after diuretics; use FEurea (<35% prerenal) when the patient is on a loop diuretic.

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13. Deeper-dives chapter rail (CH 321-330)

WHAT YOU SEE

Signpost: AKI, CKD, GN, cystic, TIN, stones, obstruction.

WHAT IT ENCODES

Roadmap of the nephrology chapters that branch off triage: 321 AKI, 322 CKD, 326 GN, 327 cystic, 328 TIN, 330 stones/obstruction. Each clue routes to a deeper chapter.

BOARD TRAP

Cystic disease (ADPKD) presents with hematuria + HTN + flank mass but is genetic/structural, not an acute nephritic GN — don't lump it with the active-sediment causes.
